ENCH296 2026 Exam - Question 6

A 10 wt% NaOH solution is made by mixing 100 g of solid NaOH (initially at 20 °C) with 900 g of water (initially at 0 °C) in a perfectly insulated vessel. The following data are provided:

Property	Value
Water heat capacity ($C_{P,water}$)	4.18 kJ kg ^{1} K ^{1}
NaOH heat capacity ($C_{P,NaOH}$)	1.46 kJ kg ^{1} K ^{1}
10 wt% NaOH solution heat capacity ($C_{P,sol}$)	3.90 kJ kg ^{1} K ^{1}
Molar mass of NaOH	40 g mol ^{1}

(a) Use an energy balance to calculate the final temperature of the mixture assuming that there is no heat of dissolution.

(b) In practice, the final temperature of the solution is measured to be 30 °C (this should be different from your prediction in part (a)). Estimate the heat of dissolution per kg of NaOH dissolved.

(c) Consider 1 kg of 10 wt% NaOH solution at 30 °C, compared with the unmixed components (100 g of solid NaOH at 30 °C and 900 g of liquid water at 30 °C). Does the solution contain more or less enthalpy than the unmixed components at the same temperature? Justify your answer.

Solution

(a) Final temperature ignoring heat of dissolution

Assumptions:

- perfectly insulated vessel ($\dot{Q} = 0$)
- no work ($\dot{W} = 0$)
- no flow — a batch system

Integrating from mixing to equilibrium gives $\Delta H_{sys} = 0$.

Energy balance approach. Each component brought from its initial temperature (T_i) to the common final temperature T_f):

$$m_{NaOH}\, C_{P,NaOH}\, (T_f - T_{NaOH,i}) + m_{water}\, C_{P,water}\, (T_f - T_{water,i}) = 0$$

Solving for T_f:

$$T_f = \frac{m_{NaOH}\, C_{P,NaOH}\, T_{NaOH,i} + m_{water}\, C_{P,water}\, T_{water,i}}{m_{NaOH}\, C_{P,NaOH} + m_{water}\, C_{P,water}}$$

$$T_f = \frac{(0.1)(1.46)(20) + (0.9)(4.18)(0)}{(0.1)(1.46) + (0.9)(4.18)} = \frac{2.92}{3.908} = 0.75\ ^\circ\mathrm{C}$$

$$T_f \approx 0.75\ ^\circ\mathrm{C}$$

(b) Heat of dissolution from the actual final temperature (30 °C)

The energy balance starts as:

$$\frac{d(m_{sys} H_{sys})}{dt} = \dot{m}_{in} H_{in} - \dot{m}_{out} H_{out} + \dot{Q} + \dot{W}$$

As there are no mass flows in or out ($\dot{m}_{in} = \dot{m}_{out} = 0$) and as the system is well insulated ($\dot{Q} = 0$) and no work is done ($\dot{W} = 0$):

$$\frac{d(m_{sys} H_{sys})}{dt} = 0$$

Separating the variables gives:

$$d(m_{sys} H_{sys}) = 0\, dt$$

Integrating between the initial and final states:

$$\int_i^f d(m_{sys}H_{sys}) = 0$$

$$m_{sys,f}H_{sys,f} - m_{sys,i}H_{sys,i} = 0$$

where $H_{sys,i}$ is the enthalpy of the unmixed components and $H_{sys,f}$ is the enthalpy of the final solution.

For this problem, it is easiest to calculate the enthalpy of the initial and final states relative to some reference state. As the water starts at 0 °C, it is convenient to use 0 °C (and 1 bar) of the *un-mixed* species as the reference state. Therefore the initial total enthalpy ($m_{sys,i}H_{sys,i}$) is:

$$m_{sys,i}H_{sys,i} = m_{NaOH}\, C_{P,NaOH}\,(T_{NaOH,i} - T_{ref}) + m_{water}\, C_{P,water}\,(T_{water,i} - T_{ref})$$

$m_{sys,f}H_{sys,f}$ is the total enthalpy of the solution (units kJ) at its actual final temperature ($T_{sol,f}$), relative to 0 °C and the *un-mixed* components:

$$m_{sys,f}H_{sys,f} = m_{sol}\, C_{P,sol}\,(T_{sol,f} - T_{ref}) + m_{sol}\,\Delta H_{diss}$$

The $m_{sol}\,\Delta H_{diss}$ is added on to account for the change from the un-mixed species at 0 °C and the mixed species at 0 °C.

Combining full terms for $m_{sys,f}H_{sys,f}$ and $m_{sys,i}H_{sys,i}$ gives:

$$\begin{aligned} m_{sol}\, C_{P,sol}\,(T_{sol,f} - T_{ref}) + m_{sol}\,\Delta H_{diss} \\ -m_{NaOH}\, C_{P,NaOH}\,(T_{NaOH,i} - T_{ref}) \\ +m_{water}\, C_{P,water}\,(T_{water,i} - T_{ref}) = 0 \end{aligned}$$

Solving for the heat of dissolution (ΔH_{diss}):

$$\begin{aligned} m_{sol}\,\Delta H_{diss} &= m_{NaOH}\, C_{P,NaOH}\,(T_{NaOH,i} - T_{ref}) \\ &\quad + m_{water}\, C_{P,water}\,(T_{water,i} - T_{ref}) \\ &\quad - m_{sol}\, C_{P,sol}\,(T_{sol,f} - T_{ref}) \end{aligned}$$

$$\begin{aligned} m_{sol}\,\Delta H_{diss} &= (0.1)(1.46)(20-0) + (0.9)(4.18)(0-0) - (1)(3.9)(30-0) \\ &= -114.08 \text{ kJ} \end{aligned}$$

$m_{sol}\,\Delta H_{diss} = -114.08$ kJ for the 100 g of NaOH dissolved. So on a mass basis of NaOH, the heat of dissolution is:

$$\Delta H_{diss} = -1141 \text{ kJ/kg}$$

The heat of dissolution is approximately **-1141 kJ per kg of NaOH dissolved**, and is exothermic.

(c) Enthalpy of the solution vs. the unmixed components

The solution contains **less** enthalpy than the unmixed components at the same temperature.

Dissolving NaOH in water is strongly exothermic — energy is released as the solid dissolves. If the 10 wt% solution were instead prepared isothermally at 30 °C, this released energy (114 kJ, from part (b)) would have to be removed from the system to hold the temperature at 30 °C.

Because that energy has left the system, the solution at 30 °C sits at a **lower** enthalpy state than the separated components at 30 °C — by approximately 114 kJ per 100 g of NaOH dissolved (1141 kJ/kg NaOH).